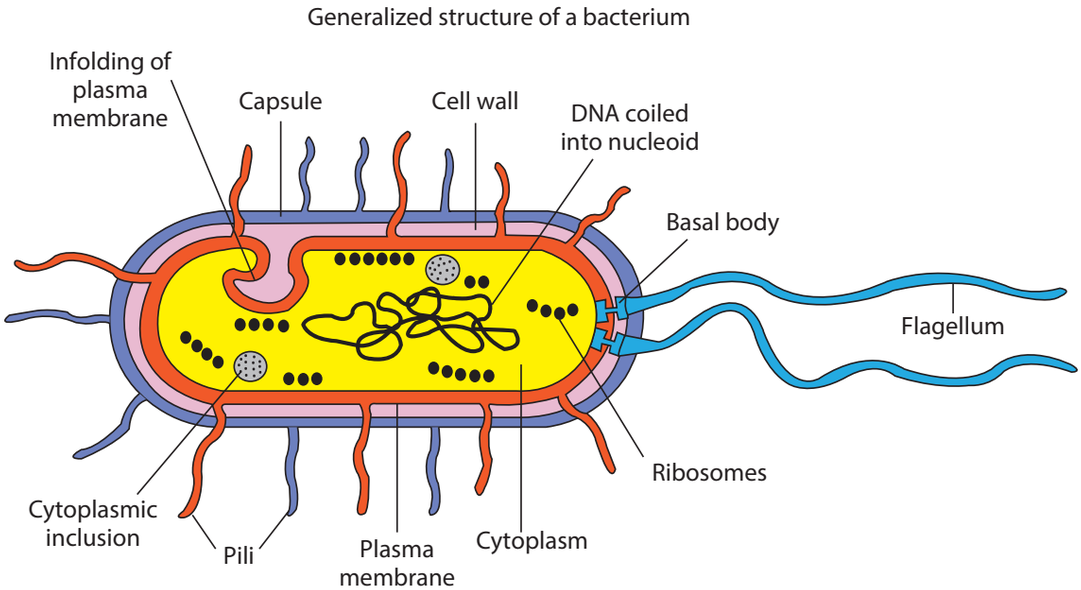


Figure 7.1: Generalized structure of a bacterium

Bacteria, cyanobacteria (blue green algae) microalgae, protozoa, yeasts and fungi represent the microorganisms. Prokaryotes are organisms with primitive type of nucleus lacking a well defined membrane (Figure 7.1). The nuclear material is a DNA molecule in prokaryotes compared to chromosomes of higher organisms. Eukaryotes are organisms with cells having true nuclei enclosed in a nuclear membrane and are structurally more complex than prokaryotes. There exists varying degree of localization of cellular functions in eukaryotes that occur in distinct membrane bound intracellular organelles like nuclei, mitochondria, chloroplasts. The cells of living organisms are either prokaryotic or eukaryotic in nature and there is not any intermediate condition. The size, shape, morphology and the internal cellular organizations are different in these two groups.

Satisfactory criteria to differentiate bacteria, fungi and algae could not be made until the development of electron microscope, which depicted the internal structure of these organisms. The absence of membrane bound internal structures in bacteria and their presence in fungi, algae, protozoa, plant and animal cells was taken as criterion to differentiate prokaryotes and eukaryotes.

7.1 Size, Shape and Arrangement of Bacteria

7.1.1 Size of Bacteria

Bacteria are minute living bodies and represent one of the lowest orders of living cells. The determination of size of the different forms is originally carried out by comparison with known RBC. A more accurate estimation is now obtained by the use of a special micrometer eye-piece, containing a graduated scale. The unit of measurement of bacteria is called micron (µ or µm). 1 micron is equal to 1 thousand of millimeter. Resolution of unaided eye is 200µm. The size of bacteria is constant but

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